

Cognitive Science — Module 03: Perception — Study Questions

1. What does it mean to say the brain is a "prediction machine"?
2. Explain the difference between top-down predictions and bottom-up sensory data.
3. Define prediction error in your own words. What triggers the brain to generate one?
4. What is Bayesian inference? Explain the three components: prior, likelihood, and posterior.
5. In the context of perception, what is a "prior"? Give an everyday example.
6. What is precision? How does it work as a "volume knob" for prediction errors?
7. How does precision change in a dark room compared to a bright room? Why?
8. Explain the Hollow Mask Illusion using the predictive processing framework.
9. Explain the McGurk Effect. What does it reveal about how the brain combines predictions?
10. What is change blindness? How does predictive processing explain why it occurs?
11. If perception is "the brain's best guess," does that mean we never see the world as it really is?
12. How is perceptual inference (updating beliefs) related to the Markov Blanket (Module 01)?
13. Why are illusions considered "features" rather than "bugs" of a predictive system?
14. Compare human perception with a camera. What does a camera do that the brain does not, and vice versa?
15. What role does context play in perception? Give an example where the same sensory data produces different perceptions depending on context.
16. How might anxiety affect perception through precision weighting? Give a concrete example.
17. What happens when the brain's prediction is completely wrong? How does it recover?
18. How does the concept of prediction error connect to "surprise" in everyday language?
19. Can you perceive something without any prior expectations? Why or why not?
20. How does the predictive processing view of perception differ from the "common sense" view that we passively receive information from the world?